

CSC 340 – PROGRAMMING METHODOLOGY

Fall 2024

Instructor:	Sara El Alaoui	Email:	selalaoui@sfsu.edu
Location:		Time:	
[01]	325 Thornton Hall		T-Th 11:00 am – 12:15 pm
[03]	106 Business Building		T-Th 12:30 pm – 1:45 pm

About CSC 340: This course, as described on the university bulletin, covers “*advanced data structures and algorithms for manipulation in C++ with an emphasis on design and implementation, practical applications, and algorithms for sorting, searching, and graphs*”. More importantly, it teaches how to critically think about a problem, how to design an efficient and effective solution, and how to professionally implement your solution in C++. These skills are of great significance for upper division courses at SFSU but also for future jobs.

Student Drop-in Hours: stop by my office (TH 930) from **2:30 pm - 3:30 pm T, 3:30 pm - 4:30 pm Th** or log in to [this Zoom link](#).

Notes 1: The Zoom meeting room will be open for drop-ins during the aforementioned time slots. I have set up a waiting room to avoid interruption to discussions in progress.

Notes 2: If these time slots do not work for you, please set up an appointment by email. We can meet in-person or via Zoom.

Prerequisites: This course builds on contents covered in CSC 220 and CSC 230. Please talk to me if you have not passed both courses with grades of C or better.

Bulletin Course Learning Objectives:

1. Students will learn and understand the concepts and applications of basic and advanced C++ topics including ADTs, pointer-based implementation of dynamic arrays and linked lists, operator overloading, STLs, inheritance, and polymorphism.
2. Students will learn the problem solving process and describe their solution to a computational problem in pseudocode.
3. Students will apply the above skills to develop medium sized C++ projects to solve real-world problems.
4. Students will learn to analyze the time complexity of foundational algorithms (recursive and non-recursive) such as binary search, insertion sort, selection sort, quick sort, and finding the k-th ranked object.

Important Resources:

- **Slides, Handouts and announcements:** all course materials and announcements will be available on Canvas. If you are having difficulty accessing and/or using Canvas, please notify me as soon as possible.
- **Discussion forum:** We will be using Canvas Discussions as needed.
- **Main Textbook:** Frank M. Carrano, *Data Abstractions and Problem Solving with C++*, 7th edition, Pearson.
- **Recommended C++ IDEs:**
 - Microsoft Visual C++ (for Windows), JetBrains CLion, NetBeans, or IDE of your choice
 - Text editor and command line using GCC (GNU Compiler Collection) or CLang compiler

Tentative Course Outline:

Week	Topics	Readings	Assignments
Week 1 Aug. 27 & 29	Course syllabus and Introduction Intro to Programming Methodology	Appendices K and B	
Week 2 Sept. 3 & 5	C++ vs. Java C++ basics	Appendices A, G, H & I	Assignment 1 Out
Week 3 Sept. 10 & 12	Problem solving Data abstraction I	Appendix B Chap. 01	
Week 4 Sept. 17 & 19	Data abstraction II Generic Programming, Templates I	Interlude 1 Chap. 03	Assignment 1 Due Assignment 2 Out
Week 5 Sept. 24 & 26	Pointers, Memory Management I Templates II	Interlude 2 Chap. 04	
Week 6 Oct. 1 & 3	Smart pointers, Memory Management II Midterm 1 Review	Other reading materials	
Week 7 Oct. 8 & 10	Loop Complexity Midterm 1	Chap. 10	Assignment 2 Due (Monday)
Week 8 Oct. 15 & 17	Inheritance, Polymorphism and BIG 5 Overloading, friends	Interlude 4 & 2 Interlude 5	. Assignment 3 Out
Week 9 Oct. 22 & 24	STL Design Decisions (Data Structures)	Interlude 7	
Week 10 Oct. 29 & 31	Recursion, Time complexity Sorting algorithms	Chap. 02 Chap. 11	. Assignment 3 Due
Week 11 Nov. 5 & 7	Trees: implementations and Algorithms	Chap. 15 Chap. 16	Assignment 4 Out
Week 12 Nov. 12 & 14	Hash maps Midterm 2 Review	Chapter 18	
Week 13 Nov. 19 & 21	Exception Handling Midterm 2	Interlude 3	Assignment 4 Due
Week 14 Nov. 26 & 28	NO CLASS		
Week 15 Dec. 3 & 5	Graphs: implementations and Algorithms	Chap. 20	Assignment 5 Out
Week 16 Dec. 10 & 12	Guest Lecture Final Exam		
Week 17 Dec. 19	NO CLASS		Assignment 5 Due

Grading Policy:

Class participation	10%
Homework assignments	50%
Midterm 1	20%
Midterm 2	20%
Final Exam	20% (optional)*

*Best two grades from midterm 1, 2 and final exam.

Grades:

	A: 93 – 100,	A-: 92.99 – 90
B+: 89.99 – 87,	B: 86.99 – 83,	B-: 82.99 – 80
C+: 79.99 – 77,	C: 76.99 – 73,	C-: 72.99 – 70
D+: 69.99 – 67,	D: 66.99 – 63,	D-: 62.99 – 60
		F 59.99 – 0

Things you need to know:

- Attending the lectures and participating in class activities are significant aspects of the course.
 - As future engineers/researchers you will be involved in team projects and discussions.
 - Working with your classmates on class activities provides you with a great opportunity to practice.
 - In-class activities will count towards your attendance.
 - For observance of religious holidays, please make sure you notify me within the first two weeks of class. Please refer to [the department of computer science policy on this](#), available [here](#).
- Cheating is not acceptable. If you need help, please ask for it, do not wait until the last minute. More on this at the end of the syllabus.
 - Assignments are to be completed individually, unless otherwise specified.
 - You are encouraged to discuss the problems with your classmates, TAs and myself, but at high level.
 - You are also encouraged to use generative AI (GenAI) tools, such as ChatGPT, **but only** as a source of extra materials and resources to enhance your learning.
 - Copying other student's code or generating code using GenAI is strictly prohibited.
 - If you discuss with other students and/or use GenAI, please make sure you mention it on your submission specifying for what and when you did so.
 - General rule: you should be able to explain, demonstrate and reproduce any code you submit.
- Assignments:
 - Assignment submission guidelines will be described on each assignment's handout.
 - Late submission are allowed for up to 48 hours with the following penalties:

within 24 hours	15 points
between 24 and 48 hours	30 points
more than 48 hours	not accepted
- You may take a make up for in-class activities if you miss them due to a medical/family emergency (documentation required) or other circumstances beyond your control. In any case, please contact me before your absence, or as soon as you are able to.

5. Communication: Please reach out to me with any questions, concerns, course feedback or if you just want to chat about your career.
- I am available in person during office hours. You are highly encouraged to set up an appointment to meet with me in person or via Zoom if you cannot make it during office hours. Email is the best way to reach me outside of class sessions and office hours.
 - **Important** – Please start all your emails subject lines with CSC340-[section number] (e.g CSC340-01 Assignment 2 question) to make sure your email gets a higher priority.
 - *You will be writing a lot professional emails in your career, so take this as an opportunity to practice.* When writing your email, please use a formal communication style: (1) use a specific subject line, (2) start with an appropriate salutation such as “Dear Dr. El Alaoui”, (2) write your questions and content in correct grammatical sentences, and (3) end the email with a traditional close and your name. For more details, please refer to [this short article](#).
 - TAs and I will do our best to respond to emails as soon as we can, but please take into consideration that we have other responsibilities. Unless it is for an emergency (this does not include assignment deadlines), please:
 - (a) allow for up to 48 hours before you send us a reminder.
 - (b) keep in mind that you might not get a response outside of working hours and during weekends.With that said, please make sure you plan your time properly and get your questions to us as early as possible. Try not to wait until a few hours before a deadline.

This is a live document, I may make changes to any of these policies and contents at any time during the semester. I will announce changes in class and post them on Canvas.

Other useful links/resources:

- For *Canvas* (<https://canvas.sfsu.edu>) support email at@sfsu.edu, or call (415)405-5555
- For *ITS Service Desk* visit <https://its.sfsu.edu/getstarted/students>
- *J. Paul Leonard Library*: Ask a librarian! Help is available in person, via chat, email, or by Zoom appointment. Details on the library’s website: <https://library.sfsu.edu/>

Disability Access: Please do reach out to me if you need any special accommodations.

More information: Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice/415-338-2472, video phone/415-335-7210) or by email (dprc@sfsu.edu).

Academic Integrity: Academic integrity, the ethical presentation of one’s own work in accordance with the rules established for this class, is required. Instances of academic misconduct will be reported to the College in which the course is housed, the Division of Graduate Studies (if a graduate student), and the Office of Student Conduct with the report being kept in those offices until a student earns his/her degree. Any instances of cheating, deceit, fabrication, forgery, plagiarism, unauthorized altering of records or submitting false documents, unauthorized collaboration, unauthorized submission of work previously given credit, or other forms of academic misconduct will be assigned a grade penalty, likely an F or a grade of zero. Failing one or more assignments or examinations for reasons of academic integrity violations may result in a final class grade of F. Students may not withdraw from classes in which they have committed academic misconduct. Consequences for violations of academic integrity may exceed an F on the assignment, examination, or class as determined by the Academic Integrity Review Committee.

Members of our academic community have a responsibility to develop an awareness of academic integrity, to cultivate skills to realize honesty in academic and community work, and to sustain actively academic honor

as a core value of our community. Students are expected to engage in behaviors that reflect well upon the university. In addition to attending to one's own actions, the Standards for Student Conduct require that students who witness academic dishonesty notify their faculty/instructor, department chair, or the Office of Student Conduct. Supporting academic integrity enhances the reputation of the University and the value attributed to degrees awarded by the University. Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation.

Disclosures of Sexual Violence (Title IX): SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the Title IX Coordinator by completing the report form available at <https://titleix.sfsu.edu>, emailing vpsaem@sfsu.edu, or calling (415) 338-2032. To disclose any such violence confidentially, contact:

- The SAFE Place – (415) 338-2208; http://www.sfsu.edu/~safe_plc/
- Counseling and Psychological Services Center – (415) 338-2208; psyservs.sfsu.edu

For more information on your rights and available resources, visit <http://titleix.sfsu.edu>.